BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 9 — Geometry and Polygons

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

General Instructions:

1. Write your name, roll number, and section clearly on the answer sheet.
2. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
3. **Section B** (Short Questions): Attempt **ALL 11** pairs, choosing the main question **OR** its alternative. Each carries **3 marks**.
4. **Section C** (Long Questions): Attempt **ALL 4** pairs, choosing the main question **OR** its alternative. Each carries **5 marks**.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	A mathematical statement assumed to be true without proof is called:	Theorem	Conjecture	Axiom	Corollary	○○○○
2	A statement believed to be true but whose truth has not yet been proved is:	Axiom	Conjecture	Theorem	Postulate	○○○○
3	The symbol used to denote similarity between two figures is:	≅	~	=	∝	○○○○
4	If two similar figures have corresponding lengths in ratio k, the ratio of their areas is:	k	k^2	k^3	2k	○○○○
5	The sum of interior angles of a polygon with *n* sides is:	(n − 1) × 180°	n × 180°	(n − 2) × 180°	(n + 2) × 180°	○○○○
6	The sum of all exterior angles of any convex polygon is always:	180°	270°	360°	540°	○○○○
7	Each interior angle of a regular hexagon measures:	108°	120°	135°	144°	○○○○
8	The area of a regular polygon with perimeter P and apothem a is:	P × a	½ P × a	¼ P × a	2P × a	○○○○
9	If the ratio of volumes of two similar solids is 27 : 8, the ratio of their corresponding lengths is:	3 : 2	9 : 4	27 : 8	6 : 4	○○○○
10	The locus of points equidistant from two given points A and B is the:	Circle through A and B	Angle bisector of AB	Perpendicular bisector of AB	Midpoint of AB only	○○○○
11	A regular polygon has exterior angle 45°. How many sides does it have?	6	8	10	12	○○○○
12	The number of diagonals in a polygon with *n* sides is:	n(n − 1)/2	n(n − 3)/2	n(n − 2)/2	n(n + 1)/2	○○○○

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	Differentiate between an **axiom** and a **conjecture** with one example of each.	**3**	**OR**	What is a **theorem**? State the six important steps used to prove a geometrical theorem.	**3**
2(ii)	Prove that **3x + 9 = 3(x + 3)** using deductive reasoning. Also verify using inductive reasoning for x = 2.	**3**	**OR**	Using the identity $(a + b)^2 = a^2 + b^2 + 2ab$, find the value of $(103)^2$ using deductive approach.	**3**
2(iii)	In ΔABC and ΔDEF, ∠A = ∠D = 40°, AC/DF = AB/DE = 2/3. Show that ΔABC ~ ΔDEF and state the condition of similarity used.	**3**	**OR**	Triangles PQR and XYZ are similar with PQ/XY = QR/YZ = PR/XZ = 2. If PQ = 6 cm, QR = 8 cm, PR = 4 cm, find the sides of ΔXYZ.	**3**
2(iv)	In the figure, ΔLMN ~ ΔXYZ. Find the values of *x* and *y* given: LM = x, LN = 6 cm, MN = 4 cm, XY = 3, XZ = y, YZ = 2.	**3**	**OR**	A pole of height 7.5 m casts a shadow of 15 m. At the same time a wall nearby casts a shadow of 45 m. Using similar triangles, find	**3**

				the height of the wall.	
2(v)	Two similar rectangles have corresponding sides in ratio 1 : 2. If the area of the smaller rectangle is 6 cm^2, find the area of the larger rectangle. Also write the general rule used.	**3**	**OR**	Two similar triangles have areas A_1 = 25 cm^2 and A_2 = 108 cm^2. A side of the smaller triangle is 2 cm. Find the corresponding side of the larger triangle.	**3**
2(vi)	Two similar cylinders have radii 3 cm and 5 cm. Find: (a) ratio of their curved surface areas; (b) ratio of their volumes.	**3**	**OR**	A model bus is built at a scale of 1 : 10 with a volume of 30 m^3. Find the volume of the actual bus.	**3**
2(vii)	Define **regular polygon** and **irregular polygon**. Give two examples of each. State the formula for each interior angle of a regular n-sided polygon.	**3**	**OR**	The exterior angle of a regular polygon is 120°. Identify the name of the polygon and find its interior angle.	**3**
2(viii)	Find the sum of all interior angles of a polygon with 9 sides. Hence find each interior angle if the polygon is regular.	**3**	**OR**	The sum of all interior angles of a polygon is 2880°. Find the number of sides and name the polygon if it is regular.	**3**
2(ix)	The exterior angles of a pentagon are (y + 5)°, (2y + 3)°, (3y + 2)°, (4y + 1)°, and (5y + 4)°. Find the value of y and each exterior angle.	**3**	**OR**	A regular polygon has each interior angle equal to five times its exterior angle. Find the number of sides of the polygon.	**3**
2(x)	A plot in a housing scheme is a regular hexagon with side 10 m. Find: (a) perimeter; (b) area, given apothem = 8.66 m.	**3**	**OR**	A lawn in the shape of an equilateral triangle has side 5 m. Find its perimeter and the cost of fencing at Rs. 220 per metre.	**3**
2(xi)	Define **locus**. Describe the locus of a point equidistant from two given intersecting straight lines and sketch it.	**3**	**OR**	A point P moves such that it is always 4 cm away from a fixed straight line l. Describe and sketch the locus of P. How many such loci exist?	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt 4 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	**(a)** Differentiate between inductive and deductive reasoning with one example each. **(b)** Prove algebraically that $(x + 1)^2 + 5 = x^2 + 2x + 6$ by taking x = 2, 5, and 10. Also prove it as a general statement using deductive approach.	**5**	**OR**	**(a)** Define: (i) mathematical statement; (ii) proposition (its three parts); (iii) corollary; (iv) converse of a theorem. Give one example each. **(b)** Which are mathematical statements? Justify. (i) 34 + 16 ≠ 50 (ii) $(a+b)^2 = a^2+2ab+b^2$ (iii) xy + z = 12	**5**
3(ii)	In ΔABC, D on AB and E on AC such that DE ∥ BC, AD = 4 cm, DB = 6 cm, BC = 8 cm. **(a)** Show that AE/EC = AD/DB. **(b)** Find AE if AC = 15 cm. **(c)** Find Area of ΔADE / Area of ΔABC using corresponding sides.	**5**	**OR**	Wajid's shadow is 2/3 of his height. A nearby pole has a shadow of 12 m. Using similar triangles: **(a)** Find the height of the pole. **(b)** If Wajid is 1.5 m tall, find the length of his shadow. **(c)** Are the triangles formed similar? State the similarity condition.	**5**
3(iii)	In ΔABC and ΔADE, vertex A is shared, DE ∥ BC, AD = 5 cm, AB = 8 cm. **(a)** Show ΔADE ~ ΔABC and state the reason. **(b)** Find the ratio DE/BC. **(c)** If Area ΔABC = 256 cm^2, find: (i) Area ΔADE; (ii) Area of trapezium DBCE.	**5**	**OR**	Two similar cylinders P and Q have radii 3 cm and 5 cm respectively. **(a)** Find the ratio of their volumes. **(b)** Volume of P is 162 cm^3. Find the volume of Q. **(c)** If curved surface area of Q is 250 cm^2, find the curved surface area of P.	**5**
3(iv)	**(a)** Find the sum of interior angles of a 13-sided polygon. Is a polygon with interior angle sum of 9 right angles possible? Give reason. **(b)** Interior angles of a hexagon: 90°, 155°, 112°, x°, 140°, 98°. Find x. **(c)** A regular polygon has exterior angle 12°. Find the number of sides and sum of interior angles.	**5**	**OR**	**(a)** Describe fully the locus of a point P equidistant from a fixed point O. Sketch it. **(b)** Two isosceles triangles XAB and YAB share base AB. Prove that line XY bisects AB perpendicularly. **(c)** Three non-collinear points A, B, C are given. Describe the locus of all points equidistant from A, B, and C. What is the special name of this point?	**5**

Invigilator's Signature: ________________ Candidate's Signature: ________________

Chapters Covered: Ch 9 — Geometry and Polygons

— End of Question Paper —